The Journal of Open Source Software

UncertRadio: Software for determining characteristic limits in accordance to DIN EN ISO 11929 for radioactivity measurements

Günter Kanisch[1], **Florian Ober**[2¶], **and Marc-Oliver Aust**[3]

1 Hamburg, Germany, formerly Thünen-Institute **2** Max Rubner-Institute, Kiel, Germany **3** Thünen-Institute, Bremerhaven, Germany ¶ Corresponding author

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Kelly Rowland

Reviewers:

- @villaa
- @alberto743

Submitted: 27 August 2024
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

DRAFT

Summary and statement of need

In Germany, radioactive substances in the environment are monitored in accordance with the Treaty establishing the European Atomic Energy Community (EURATOM) of 1957 and the Radiation Protection Act (*Gesetz Zum Schutz Vor Der Schädlichen Wirkung Ionisierender Strahlung*, 2017). The Evaluation of radiometric measurements requires the estimation of associated uncertainties as defined in the ISO GUM (Joint Committee for Guides in Metrology, 2008b). In addition, German law requires that the `characteristic limits` (decision threshold (DT) and the detection limit (DL)) are determined on the basis of this uncertainty in accordance with *DIN EN ISO 11929-1:2021-11* (2021) to *DIN EN ISO 11929-4:2021-06* (2021).

`UncertRadio` is the only publicly available software to determine these values in a user-centralized way for a variety of applications from Alpha, Beta and Gamma radiation measurements including dosimetry for up to three radionuclides simultaneously. Therefore, it is especially suited for modern liquid scintillation measurement procedures of e. g. Strontium isotopes. The user only needs to define the evaluation model by providing a set of equations to calculate the output quantity value. However, the required partial derivatives are calculated internally and don't need to be supplied by the user.

There are two main analytical approaches used within the software:

- Procedures without linear unfolding: The basic evaluation model is linear in the net count rate. Thus, the output value can be calculated directly (Kanisch, 2016a).
- Procedures utilizing linear unfolding methods: The model additionally includes linear least squares procedures for fitting e. g. time-dependent decay or build-up curves (Kanisch, 2016b).

Additionally, the model can be evaluated with a Monte Carlo simulation (*DIN EN ISO 11929-2:2021-11*, 2021) (Figure 1). This represents the method of propagating whole distributions, which has advantages if the distributions of the input variable distributions significantly deviate from the normal distribution; see ISO GUM Supplements 1 and 2 Joint Committee for Guides in Metrology (2011).

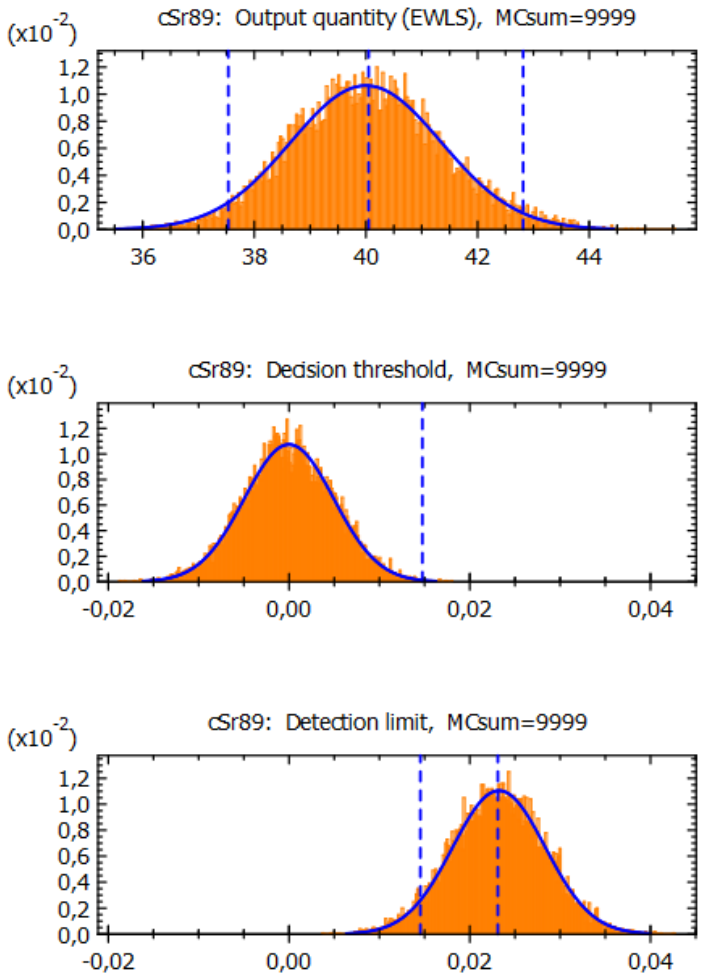

Figure 1: Example of results obtained with the Monte Carlo simulation (*DIN EN ISO 11929-2:2021-11*, 2021). The results (output quantity, decision threshold and detection limit) calculated following *DIN EN ISO 11929-1:2021-11* (2021) are also included.

Scientific references

`UncertRadio` has been used in several scientific publications. It is referenced by the standard *DIN EN ISO 11929-1:2021-11* (2021) to *DIN EN ISO 11929-4:2021-06* (2021) series and actively used by the German authorities for monitoring environmental radioactivity and external radiation.

In Kanisch (2016a), an overview considering evaluations without using linear unfolding has been presented. Two significant linear relationships in the model equations for the net count rate (common in evaluation models) were identified providing a generalized approach for the determination of the characteristic limits. The second part (Kanisch, 2016b) extends the evaluation models to include linear unfolding methods utilizing a (weighted) linear least-squares approach for the first stage of the model. This step is solved using matrix-algebra which also takes parameters with uncertainties in the design matrix into account.

Applications, examples and quality control

UncertRadio includes a set of approximately 70 example projects, which are structured text files available in both English and German language. They provide an impression of how the set of equations is organized for various measurement models. A short overview of the example projects is given in section 3.3 of the UncertRadio Help file. These examples contributed to the validation of UncertRadio. To check if UncertRadio is working correctly, all examples can be run automatically by selecting "Options/QC batch test" in the main menu.

Availability and documentation

UncertRadio is available free for download as compiled windows binaries since 2014.

Recently, it was decided to make the source code available as open source software under the GNU General Public License 3. UncertRadio is written in modern Fortran utilizing many Fortran 2003 and 2008 features, e. g. the C-interoperability.

The graphical user interface is build with GTK 3 in combination with gtk-fortran (Magnin et al., 2019), which provides the required Fortran bindings. PLplot is implemented for the graphical presentations (Alan W. Irwin et al., 2019).

Many of the utilized numerical procedures are derived from the work of Alan Miller, John Burkardt and Brandt (1999). A function parser (Schmehl, 2008) is included for interpreting user-defined equations.

The UncertRadio source code is available on GitHub. Detailed building instructions are provided within the README file. UncertRadio works both on Linux and Windows and comes with language packages for English, French and German. Nevertheless, only Windows binaries are currently provided for download. They are available on GitHub and on the homepage of the Thünen-Institute.

UncertRadio contains an extensive compiled HTML help (chm) file for the description of the program features in German and English language. The individual help topics are available from within UncertRadio using various help buttons. A sphinx documentation based on this help file is in preparation. The current development goals and open issues can be found in the README file.

Code problems can be reported in the issues tab on GitHub. The authors are happy to help within their capabilities. Feedback and contributions via pull request are greatly appreciated.

Conflict of interest

The authors declare no financial conflicts of interest.

Acknowledgements

The authors thank all those who have contributed to improving UncertRadio by reporting problems and suggesting new features. We also thank D. Schrammel (KIT, Karlsruhe, Germany) for many comparison test calculations and Prof. R. Michel (formerly IRS, Hannover, Germany), who tested and applied UncertRadio during preparation of *DIN EN ISO 11929-4:2021-06* (2021).

Furthermore, the authors especially thank Marco Lechner and Josef Peter, who have contributed with their knowledge and constructive suggestions to make UncertRadio open source.

DRAFT

References

Alan W. Irwin, Andrew Ross, Geoffrey Furnish, Hazen Babcock, António Tomé, Arjen Markus, Andrew Roach, Hezekiah M. Carty, Doug Hunt, James Dishaw, Jerry Bauck, Maurice LeBrun, Phil Rosenberg, & Werner Smekal. (2019). *PLplot*. http://plplot.org/index.php

Brandt, S. (1999). *Datenanalyse: Mit Statistischen Methoden Und Computerprogrammen*. Spektrum Akademischer Verlag. ISBN: 978-3-8274-0158-8

DIN EN ISO 11929-1:2021-11: Bestimmung der charakteristischen Grenzen (Erkennungsgrenze, Nachweisgrenze und Grenzen des Überdeckungsintervalls) bei Messungen ionisierender Strahlung - Grundlagen und Anwendungen - Teil 1: Elementare Anwendungen (DIN ISO). (2021). Beuth Verlag GmbH. https://www.beuth.de/de/norm/din-en-iso-11929-1/337873649

DIN EN ISO 11929-2:2021-11: Bestimmung der charakteristischen Grenzen (Erkennungsgrenze, Nachweisgrenze und Grenzen des Überdeckungsintervalls) bei Messungen ionisierender Strahlung - Grundlagen und Anwendungen - Teil 2: Fortgeschrittene Anwendungen. (2021). Beuth Verlag GmbH. https://www.beuth.de/de/norm/din-en-iso-11929-2/337873700

DIN EN ISO 11929-4:2021-06: Bestimmung der charakteristischen Grenzen (Erkennungsgrenze, Nachweisgrenze und Grenzen des Überdeckungsintervalls) bei Messungen ionisierender Strahlung - Grundlagen und Anwendungen - Teil 4: Anwendungsleitfaden (ISO 11929-4:2020). (2021). Beuth Verlag GmbH. https://www.beuth.de/de/norm/din-iso-11929-4/336037047

Gesetz zum Schutz vor der schädlichen Wirkung ionisierender Strahlung: StrlSchG. (2017). https://www.gesetze-im-internet.de/strlschg/

Joint Committee for Guides in Metrology. (2008a). *Evaluation of measurement data - Supplement 1 to the GUM: JCGM 101:2008*. International Bureau of Weights and Measures. https://www.bipm.org/en/publications/guides/gum.html

Joint Committee for Guides in Metrology. (2008b). *Guide to the expression of uncertainty in measurement: JCGM 100:2008*. International Bureau of Weights and Measures. https://www.bipm.org/en/publications/guides/gum.html

Joint Committee for Guides in Metrology. (2011). *Evaluation of measurement data - Supplement 2 to the GUM: JCGM 102:2011*. International Bureau of Weights and Measures. https://www.bipm.org/en/publications/guides/gum.html

Kanisch, G. (2016a). Generalized evaluation of environmental radioactivity measurements with UncertRadio Part I: Methods without linear unfolding. *Applied Radiation and Isotopes*, *110*, 28–41. https://doi.org/10.1016/j.apradiso.2015.12.003

Kanisch, G. (2016b). Generalized evaluation of environmental radioactivity measurements with UncertRadio Part II: Methods with linear unfolding. *Applied Radiation and Isotopes*, *110*, 74–86. https://doi.org/10.1016/j.apradiso.2015.12.046

Magnin, V., Tappin, J., Hunger, J., & Lisle, J. D. (2019). Gtk-fortran: A GTK+ binding to build Graphical User Interfaces in Fortran. *Journal of Open Source Software*, *4*(34), 1109. https://doi.org/10.21105/joss.01109

Schmehl, R. (2008). *Fortran 95 function parser*. https://fparser.sourceforge.net/

DRAFT